BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

LESSON 014 · Linear Search and Algorithm Tracing

Name: _____ Class: _____ Date: _____

Time: _____ Teacher: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does sequence mean in programming?

Q2. 40 images are 200 KB each. Using decimal units, what is their total size in MB?
A. 0.24 MB
B. 8 MB
C. 0.2 MB
D. 800 MB

Q3. What should guide the choice of a storage device?

Q4. A laptop's SSD and screen work, but its CPU cannot run instructions. Why can it not work out a total from a saved program?

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I Do · Read new knowledge

Key vocabulary

linear search — Checks items one by one for a target.

target — Value being searched for.

index — Position used to access an array item, starting at zero.

comparison — Checking an item against a value or condition.

New learning

A sequential scan: Linear search starts at one end of a collection and compares one element at a time with the target.

Zero-based indices: The first array element is at index 0, then 1, 2, etc. Do not confuse index with value.

Stopping: If the target is found, report its index. If the list ends without a match, report not found.

One strength: Linear search works even if values are unsorted, though it may need to inspect the entire list.

Worked example

Trace linear search for 8 in [3, 9, 1, 8, 5]. Give the comparisons and index.

1. Start at index 0 and compare 3 with 8: no match.

2. Index 1: 9 is not 8.

3. Index 2: 1 is not 8.

4. Index 3: 8 matches the target.

5. Report found at index 3 after four comparisons.

We Do · Practise with guidance

Guided 1

Search [7, 2, 6, 4] for 6. Which elements are compared?

Worked method: 7, 2, then 6. Found at index 2 after three comparisons.

Guided 2

Search [7, 2, 6, 4] for 9. What should be returned?

Worked method: Not found, after checking all four elements.

Guided 3

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Must this array be sorted for linear search? Explain.

Worked method: No. Linear search works with unsorted values because it checks sequentially.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Code First · Predict → Write → Test → Improve

Write one line to display the PC register name.

```
registerName = "PC"
```

write

Question 1

State whether linear search requires a sorted list. [1 mark]

Question 2

Search [9, 4, 1, 6] for 1. State each comparison and the zero-based index returned. [3 marks]

Question 3

Explain how linear search would find 7 in an unsorted list [3, 8, 7, 1], or report failure. [4 marks]

Exit Ticket · Answer before checking

Linear search can only work when a list is already sorted.

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Instructions execute in the specified order.

Q2. B — 8 MB

Q3. The application's requirements, including capacity, speed, cost and physical use.

Q4. An SSD stores the program. The CPU must fetch, decode and execute its instructions to process the numbers.

I Do / We Do pages · Guided practice

Q1. 7, 2, then 6. Found at index 2 after three comparisons.

Q2. Not found, after checking all four elements.

Q3. No. Linear search works with unsorted values because it checks sequentially.

You Do pages · Code First

```
registerName = "PC"
print(registerName)
```

Expected result: PC

You Do pages · Independent questions

Q1. No. Linear search can examine items in an unsorted list [1].

Q2. Compare 9 [1], 4 [1], then 1 at index 2 [1].

Q3. Start at first element [1]; compare sequentially 3, 8, 7 [1]; report index 2 when found [1]; if end reached with no match report not found [1].

Exit Ticket · Model answer

False

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

BLACK PEN Answer all questions first.

GREEN PEN Then: tick correct answers; rewrite mistakes.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: